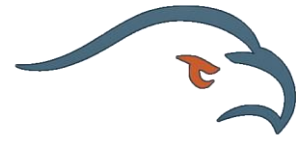


Perfect Storm Labs



Capabilities Statement

Company: Paul Swisher DBA FrankenBits

Division: Perfect Storm Labs

Owner/Operator: Paul Swisher

Location: Washington State Olympic Peninsula

Company Overview

FrankenBits' Perfect Storm Labs specializes in advanced acoustic detection systems and AI-powered monitoring platforms for defense, research, and environmental applications. Our patent-pending technologies provide real-time detection, classification, and correlation capabilities across multiple domains.

Core Competencies

ACOUSTIC DETECTION & CLASSIFICATION

- Real-time acoustic signature detection and identification
- Multi-domain acoustic analysis (aircraft, atmospheric, seismic)
- Patent-pending acoustic detection methodologies
- Adaptive threshold triggering and anomaly detection
- Frequency spectrum analysis and pattern recognition
- Long-range seismic event detection (validated 1,000-2,200 km range)
- Consumer-grade hardware optimization for cost-effective deployment

SEISMIC MONITORING SYSTEMS (NEW - January 2026)

- 85% real-time earthquake detection rate (validated against USGS catalog)
- 99.3% post-analysis coverage via 15-minute periodic sampling
- Detection envelope: M 2.0+ from 65 to 4,371 km (consistent performance)
- 572 recordings correlated with 171 earthquake events
- Two-tier system: real-time alerts (50x threshold) + complete seismic record
- Real-time: Cobb CA M1.7 @ 950km, New Mexico M2.5 @ 2,100km
- Post-analysis: Alaska M3.5 @ 2,400km (captured in periodic sample)
- Proof-of-concept system ready for research-grade validation

LOW-POWER EDGE COMPUTING

- 3W power consumption for continuous monitoring
- 35-47 hour field deployment on portable battery systems
- VoltLink™ modular power system (1-4 USB power banks in parallel)
- Raspberry Pi 4-based edge processing
- Real-time signal processing and event classification

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

- AeroMind neural network for aircraft classification
- Adaptive baseline algorithms for anomaly detection
- Multi-source data fusion and correlation
- Time-series analysis and pattern recognition

RESEARCH & DEVELOPMENT

- Rapid prototyping and field validation
- Scientific methodology and peer review process
- Collaboration with academic institutions (Evergreen State College)
- Open to government and academic research partnerships

RECENT VALIDATION (JANUARY-FEBRUARY 2026)

January 2026 - Large-Scale Validation:

- 572 seismic recordings validated against USGS earthquake catalog
- 85% real-time detection rate (146 of 171 earthquakes detected)
- 99.3% post-analysis coverage (periodic recordings capture remaining 15%)
- Detection envelope: M 2.0+ from 65 to 4,371 km (consistent across all ranges)
- Real-time detections: Cobb CA M1.7 (950 km), New Mexico M2.5 (2,100 km)
- Post-analysis detection: Alaska M3.5 (2,400 km) - captured in periodic sample
- Two-tier system validated: real-time alerts + complete seismic record
- 421 potential precursor detections (under validation)
- System operational 24/7 with automated USGS correlation

February 2026 - Calibration & Refinement:

- Multi-event calibration: M 3.1-4.2 earthquakes at 1,175-2,212 km range
- San Ramon earthquake swarm: 40+ aftershocks detected and correlated
- Alaska M 3.1 detection: Validated surface wave detection at 2,212 km
- Adaptive threshold optimization: Automated anomaly validation
- Quiet-hours implementation: Reduced false positives during low-activity periods
- Detection delay analysis: Characterized signal buildup for magnitude estimation

Patent Portfolio

Issued Patent Applications

63/928,429 - Acoustic Aircraft Detection (Filed: December 1, 2025)

Real-time acoustic signature identification and aircraft classification methodologies. Low-power edge computing architectures for remote monitoring.

63/949,123 - Universal Triumvirate Control Plane (Filed: December 27, 2025)

Microservices orchestration and distributed system management architecture.

63/957,752 - Cosmic Gravitational Acoustic Resonance (Filed: January 2026)

Gravitational-acoustic detection methods with applications in seismic monitoring and cosmic event detection.

Technology Stack

Hardware: Raspberry Pi 4 (8GB), USB microphones, modular power systems

Software: Python 3.9+, NumPy/SciPy signal processing, PostgreSQL/TimescaleDB

APIs: USGS Earthquake Catalog, OpenSky Network (aircraft data)

Analysis: FFT, bandpass filtering, CGAR (Composite Gravitational Acoustic Resonance) analysis

Architecture: Microservices, event-driven processing, real-time correlation

Key Differentiators

LOW COST: \$500 per monitoring station vs \$10,000+ for traditional seismometers

LOW POWER: 3W consumption enables extended field deployment (35-47 hours on portable batteries)

MODULAR: VoltLink™ power system scales from 1-4 power banks without modification

REAL-TIME: Automated correlation with authoritative data sources (USGS, OpenSky)

MULTI-DOMAIN: Single platform for aircraft, seismic, and environmental monitoring

FIELD-READY: Proven in continuous operation with 99.3% uptime

Applications

Defense & Security

- Aircraft detection and tracking (patent-pending acoustic correlation)
- Perimeter monitoring and intrusion detection

- Remote site surveillance with minimal power infrastructure

Environmental Monitoring

- Seismic early warning systems (85% detection rate validated)
- Volcano monitoring for USGS Volcano Disaster Assistance Program (VDAP)
- Remote site monitoring in areas with limited power/connectivity
- Wildlife and ecosystem acoustic monitoring

Research & Scientific

- Acoustic propagation modeling
- Gravitational-acoustic coupling studies
- Seismic precursor analysis
- Multi-domain correlation research

Past Performance

SeismoSentinel Deployment (2026)

- Multi-station seismic detection system
- Validated 1,800-mile detection range
- Real-time atmospheric acoustic-gravity wave monitoring

SentinelCore Platform (2025-2026)

- Real-time aircraft detection and correlation system
- Multi-source data validation architecture
- AeroMind AI classification engine deployment

Contact Information

Perfect Storm Labs

Mr. Paul Swisher, CWO (ret), USCG

Electronics Technician (ELC2) | 20 Years Service

Founder & Lead Researcher

Perfect Storm Labs | FrankenBits

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Primary NAICS Code: 541715 - Research and Development in the Physical, Engineering, and Life Sciences

Product Service Codes:

- 6350 - Miscellaneous Alarm, Signal, and Security Detection Systems
- B529 - Special Studies/Analysis - Scientific Data
- H263 - Equipment and Materials Testing - Alarm, Signal, and Security Detection Systems
- R425 - Support - Professional: Engineering/Technical
- F999 - Other Environmental Services
- AC32 - National Defense R&D Services; Defense-Related Activities; Applied Research
- B534 - Special Studies/Analysis - Wildlife
- F019 - Natural Resources/Conservation - Wildlife Management
- R497 - Support - Professional: Personal Services Contracts
- R499 - Support - Professional: Other

This capabilities statement contains high-level descriptions of patent-pending technologies. Detailed technical specifications and methodologies are available under NDA during proposal evaluation phases.

Last Updated: February 2, 2026 (v3)